



Oral PCSK9 Inhibitor Enlicitide versus Oral Non-statin Therapies: A Phase 3 Randomized Clinical Trial

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Enlicitide versus Oral Non-statin Therapies

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Trial Registration: ClinicalTrials.gov: NCT06450366

Tweet: Oral enlicotide lowered LDL-C, ApoB, & non-HDL-C more than oral non-statin comparators at day 56 in statin-treated adults with a history of a major ASCVD event or at increased risk for a first event. #JACC #ASCVD #enlicotide

Abstract

Background: Many adults fail to achieve guideline-directed low-density lipoprotein cholesterol (LDL-C) goals on statin monotherapy, requiring additional non-statin lipid-lowering medication. Enlicotide, an oral PCSK9 inhibitor, lowered LDL-C by 60% compared with placebo; its efficacy compared with other, oral non-statin therapies has yet to be examined.

Objectives: This study assessed the efficacy of enlicotide, a novel oral PCSK9 inhibitor, versus oral non-statin therapies.

Methods: In this phase 3, randomized, double-blind, active-comparator trial, statin-treated adults aged ≥ 18 years with LDL-C ≥ 55 mg/dL and a prior major ASCVD event or LDL-C ≥ 70 mg/dL if at intermediate to high risk for a first event were randomized in 2:1:1:2 fashion to enlicotide 20 mg (n = 101), bempedoic acid 180 mg (n = 50), ezetimibe 10 mg (n = 50), or bempedoic acid 180 mg plus ezetimibe 10 mg (n = 100) once daily for 56 days. The primary endpoint was mean percentage change in LDL-C from baseline to day 56; secondary endpoints included mean percentage changes in apolipoprotein B (ApoB) and non-high-density lipoprotein cholesterol (non-HDL-C). Safety endpoints included overall adverse events (AE) and discontinuations due to AEs

Results: Among 301 randomized participants (mean age, 64.4 years; 37% women; 98% receiving moderate/high-intensity statin), 298 (99.0%) completed the trial. The mean percentage change from baseline in LDL-C to day 56 was -64.6% (95% confidence interval [CI]: -68.3 to -60.9) with enlicotide, -6.3% (95% CI: -13.5 to 0.8) with bempedoic acid, -27.8% (95% CI: -32.3 to -23.4) with ezetimibe, and -36.5% (95% CI: -40.8 to -32.2) with bempedoic acid plus ezetimibe; enlicotide was superior to each comparator (all $P < 0.001$). Reductions in ApoB and non-HDL-C were also greater

with enlicitide (all $P < 0.001$). Proportions of participants with AEs and discontinuations due to AEs were similar across treatment arms.

Conclusions: In statin-treated adults with a history of a major ASCVD event or at increased risk for a first event, enlicitide achieved greater reductions in LDL-C, ApoB, and non-HDL-C than other oral non-statin therapies, demonstrating its potential role as an important add-on option when LDL-C goals are not met with statins alone.

Trial Registration: ClinicalTrials.gov: NCT06450366

Keywords: Atherosclerotic cardiovascular disease; enlicitide; PCSK9; low-density lipoprotein cholesterol; apolipoprotein B, non-high-density lipoprotein cholesterol.

ApoB = apolipoprotein B; ASCVD = Atherosclerotic cardiovascular disease; CV: cardiovascular; HeFH = heterozygous familial hypercholesterolemia; Lp(a) = Lipoprotein(a); LDL-C = low-density lipoprotein cholesterol; mAb: monoclonal antibody; non-HDL-C = non-high-density lipoprotein cholesterol; PCSK9 = proprotein convertase subtilisin/kexin type 9; PCSK9i: inhibitor of proprotein convertase subtilisin/kexin type 9 inhibitor.

Condensed abstract: In this phase 3, randomized, double-blind, active-comparator trial, statin-treated adults with a history of a major ASCVD event or at increased risk for a first event (N = 301; 298 completed) received oral PCSK9 inhibitor enlicitide 20 mg, bempedoic acid 180 mg, ezetimibe 10 mg, or co-administered bempedoic acid 180 mg plus ezetimibe 10 mg for 56 days. Mean percentage change from baseline in LDL-C to day 56 was -64.6% with enlicitide versus -6.3%, -27.8%, and -36.5%, respectively (all $P < 0.001$). ApoB and non-HDL-C also decreased more with enlicitide versus each comparator (all $P < 0.001$).

Perspectives

Clinical Competencies

Competency in Medical Knowledge: In statin-treated adults with a history of a major atherosclerotic cardiovascular disease event or at increased risk for a first event, the oral proprotein convertase subtilisin/kexin type 9 (PCSK9) inhibitor enlicitide achieved greater reductions in LDL-C, ApoB, and non-HDL-C versus bempedoic acid, ezetimibe, and bempedoic acid plus ezetimibe.

Competency in Patient Care: Enlicitide may offer a highly effective oral option for patients who require additional LDL-C lowering beyond statins to reach lipid goals.

Translational Outlook 1: Broader use of effective oral LDL-C-lowering therapies could help address the global burden of the cardiovascular disease epidemic.

Introduction

Atherosclerotic cardiovascular disease (ASCVD) remains a leading cause of morbidity and mortality despite widespread availability of statins and other lipid-lowering therapies.¹⁻³ Lowering low-density lipoprotein cholesterol (LDL-C) has been consistently shown to reduce the risk of ASCVD events, with the magnitude of risk reduction directly related to both the absolute degree and the duration of LDL-C lowering.⁴ Accordingly, current guidelines recommend intensive LDL-C reduction,^{5,6} particularly among individuals with established ASCVD;⁷ however, many patients receiving statin therapies do not achieve recommended LDL-C goals.^{8,9}

Inhibition of proprotein convertase subtilisin/kexin type 9 (PCSK9) offers an effective strategy for further reducing LDL-C levels. Injectable PCSK9 monoclonal antibodies (mAb) lower LDL-C by approximately 60% beyond statin therapy and have been shown to reduce major adverse cardiovascular (CV) events,¹⁰⁻¹³ but their uptake remains limited by injection requirements, cost, and access barriers.^{9,14,15} Closing the persistent LDL-C treatment gap represents an opportunity to meaningfully alter the course of the CV epidemic, in which preventable ASCVD events remain the leading cause of death despite widespread statin use.^{2,16-18} Thus, an effective oral PCSK9 inhibitor (PCSK9i) could address an important unmet need in lipid management.

Oral non-statin therapies—including bempedoic acid, ezetimibe, and the combination of bempedoic acid and ezetimibe—can provide additional LDL-C lowering when added to statins, although the magnitude of the effect is modest compared with injectable PCSK9is.^{11,13,19}

Understanding the differences in efficacy between add-on therapies can be helpful to inform shared decision making between clinicians and patients regarding optimal treatment selection.

Enlicitide decanoate (enlicitide) is an oral macrocyclic peptide PCSK9i that blocks PCSK9 binding to the LDL receptor via the same mechanism as injectable mAbs, ultimately reducing

LDL receptor degradation and enhancing LDL-C clearance.^{20,21} In two phase 3, randomized, placebo-controlled trials, enlicitide reduced LDL-C levels by approximately 60% compared with placebo when added to statin therapy and was well tolerated.^{20,21} Enlicitide also led to reductions in other atherogenic particles, including apolipoprotein B (ApoB), non-high-density lipoprotein cholesterol (non-HDL-C), and lipoprotein(a) (Lp(a)), and enabled a high proportion of patients to achieve guideline-recommend LDL-C goals.^{20,21}

The phase 3 CORALreef AddOn trial compared the efficacy of enlicitide versus bempedoic acid, ezetimibe, and bempedoic acid plus ezetimibe, when added to a background of statin therapy, in adults at risk for major ASCVD events.

Methods

This phase 3 clinical trial was conducted in accordance with applicable regulatory requirements and the principles of Good Clinical Practice. For all participating sites, the study protocol (**Supplement 1**) and all amendments were approved by an independent ethics committee. All participants provided written informed consent. This report was prepared in accordance with the Consolidated Standards of Reporting Trials (CONSORT) reporting guideline.²² Data were analyzed by statisticians employed by the study sponsor and were interpreted by all authors, including the academic authors and representatives of the study sponsor. The sponsor provided results and methodological details and all authors participated in the preparation, review, and approval of the manuscript. The decision to publish was made jointly by all authors who vouch for the fidelity of the trial and analyses to the trial protocol and statistical analysis plan.

Study Design

This was a phase 3, double-blind, randomized, active-comparator, parallel-group trial (Protocol number MK-0616-018) conducted from July 8, 2024 to March 28, 2025, at 35 sites in 8 countries. After a screening period of up to 30 days, eligible participants were randomized in a 2:1:1:2 fashion to receive enlicitide 20 mg, bempedoic acid 180 mg, ezetimibe 10 mg, or co-administered bempedoic acid 180 mg and ezetimibe 10 mg, once daily for 56 days (**Supplementary Figure 1**). After day 56, participants either enrolled in an open-label extension study or completed a safety follow-up telephone visit approximately 56 days after the last dose (day 112).

Randomization and Masking

Randomization occurred centrally with an interactive response technology system using a computer-generated allocation sequence, stratified by estimated glomerular filtration rate (eGFR; <60 versus ≥ 60 mL/min/1.73 m²).

Matching placebos, corresponding to enlicitide, bempedoic acid, and ezetimibe, were used to maintain blinding of the active interventions. Enlicitide contained the excipient sodium caprate, a permeation enhancer; matching placebo for enlicitide did not contain sodium caprate.

Participants, investigators, site staff, and sponsor personnel remained blinded throughout the study.

Study Population

Eligible participants were aged 18 years or older with hypercholesterolemia who either had a history of a major ASCVD event and an LDL-C level ≥ 55 mg/dL (1.42 mmol/L), or who were at intermediate to high risk for a first major ASCVD event and had an LDL-C level of ≥ 70 mg/dL (1.81 mmol/L) at screening.

All participants were receiving background statin therapy (low-, moderate- or high-intensity). Concomitant use of ezetimibe, bempedoic acid, or commercially available PCSK9is and certain medications with potential for interaction with one of the study interventions was prohibited (**Supplementary Table 2**).

Key exclusion criteria included a diagnosis of homozygous²³ or compound or double heterozygous familial hypercholesterolemia (FH); uncontrolled diabetes mellitus (hemoglobin A1c $\geq 9\%$); a major ASCVD event within 3 months before screening; planned arterial revascularization within 8 weeks after randomization; severe renal insufficiency (estimated glomerular filtration rate <30 mL/min/1.73 m²) or dialysis; clinically significant hepatic disease or liver test abnormalities; fasting triglyceride levels of 400 mg/dL (4.52 mmol/L) or greater; a history of tendon disorder, tendon rupture, or gout; and current or recent treatment with an injectable PCSK9i, angiopoietin-like protein 3 inhibitor, or microsomal triglyceride transfer protein inhibitor at Screening. Additional eligibility criteria are provided in **Supplement 1**.

Study Procedures

Participants were instructed to take the study intervention once daily on an empty stomach in the morning and to not consume food or beverages (other than water) until 30 minutes after dosing. Adherence with the study intervention (dose compliance; calculated as the percentage of expected doses taken [number dispensed – number returned]/number expected x 100) was assessed by counting returned pills at each visit and corroborated by participant reporting. Adherence to administration instructions was assessed separately by participant self-report at study visits. Background lipid-lowering therapy was continued at stable doses throughout the trial. Participants in all treatment arms received the relevant active treatment(s) and matching placebo for the other

treatment(s) to maintain the blind. Each participant was instructed to take 3 pills each day (2 tablets and 1 capsule). Those in a single active drug (enlicotide, ezetimibe, or bempedoic acid) treatment group took 1 active pill and 2 placebo pills each day. Those in the co-administered bempedoic acid and ezetimibe group took 1 bempedoic acid capsule, 1 ezetimibe tablet, and 1 enlicotide placebo tablet each day.

Study visits occurred at screening, baseline (day 1), day 21, day 42, and day 56, with telephone contact on day 14. Participants who discontinued study intervention were encouraged to complete all remaining scheduled assessments. Laboratory evaluations were conducted centrally, and efficacy assessments remained masked through study completion. Investigators were instructed not to perform local efficacy testing.

Assessments

Fasting lipid panels were obtained at screening and on days 1, 21, and 56, and at treatment-discontinuation visits. ApoB and Lp(a) were measured on days 1, 21, and 56, and at treatment-discontinuation visits.

Throughout the trial, LDL-C was calculated using the Martin-Hopkins equation. At baseline, LDL-C levels were also assessed by β -quantification in all participants. During follow-up, β -quantification measurements were obtained if the calculated LDL-C level was 40 mg/dL (1.03 mmol/L) or less, if data were missing, or if the triglyceride levels were 400 mg/dL (4.52 mmol/L) or greater. If the LDL-C level was measured by β -quantification at day 56, the baseline β -quantification-derived LDL-C value was used in analyses to ensure that changes in LDL-C levels were assessed with the use of the same method within a participant when possible.

Safety assessments included adverse events (AEs), vital signs, physical examinations, and clinical laboratory testing. AEs were collected from the time of randomization through study completion, defined as enrolment in the open-label extension study or completion of the safety follow-up telephone visit approximately 56 days after the last dose. New-onset or worsening diabetes mellitus and potential drug-induced liver injury were prespecified events of clinical interest. An independent external data-monitoring committee reviewed unblinded safety data.

Study Endpoints

The primary endpoint was the mean percentage change from baseline in LDL-C to day 56. Key secondary endpoints (multiplicity controlled) included the mean percentage change from baseline in ApoB and non-HDL-C to day 56. The multiplicity adjustment was reserved for these primary and key secondary endpoints that directly support the primary objective of the study and intended for confirmatory inference. Additional prespecified secondary endpoints not controlled for multiplicity were included for supportive evidence to understand the broader clinical profile of enlicitide. These endpoints included the percentage change from baseline in Lp(a) to day 56 and the proportion of participants achieving prespecified LDL-C targets at day 56. Safety endpoints included the proportion of participants with AEs and the proportion who discontinued study intervention due to AEs.

Multiplicity

Testing was performed in order of the primary and key secondary efficacy endpoints with hypotheses as below and was to be stopped with the first comparison that had a one-sided p-value ≥ 0.025 : H1: enlicitide is superior to bempedoic acid on mean percent change from baseline in LDL-C to Day 56; H2: enlicitide is superior to ezetimibe on mean percent change from baseline in LDL-C to Day 56; H3: enlicitide is superior to bempedoic acid plus ezetimibe on mean percent

change from baseline in LDL-C to at Day 56; H4: enlicitide is superior to bempedoic acid on mean percent change from baseline in ApoB to at Day 56; H5: enlicitide is superior to ezetimibe on mean percent change from baseline in ApoB to at Day 56; H6: enlicitide is superior to bempedoic acid on mean percent change from baseline in non-HDL-C to at Day 56; H7: enlicitide is superior to ezetimibe on percent change from baseline in non-HDL-C to at Day 56; H8: enlicitide is superior to bempedoic acid plus ezetimibe on mean percent change from baseline in ApoB to at Day 56; H9: enlicitide is superior to bempedoic acid plus ezetimibe on mean percent change from baseline in non-HDL-C to at Day 56.

Statistical Analysis

Assuming a treatment difference with respect to mean percent change from baseline in LDL-C to Day 56 for enlicitide minus bempedoic acid, enlicitide minus ezetimibe and enlicitide minus bempedoic acid plus ezetimibe of -33%, -27% and -12%, respectively, a planned sample size of 300 participants provided greater than 99% power to demonstrate superiority versus bempedoic acid or ezetimibe, and approximately 91% power versus bempedoic acid plus ezetimibe, using a one-sided α of 0.025. Calculations accounted for anticipated imputation-related attenuation and approximately 3% missing data at day 56.

Efficacy analyses were conducted in the full analysis set, including all randomized participants who received at least one dose of double-blind study intervention. Safety analyses included all randomized participants who received at least one dose of double-blind study intervention and were analyzed according to the treatment received.

For each pairwise comparison, the multiplicity-controlled endpoints were analyzed using analysis of covariance models with treatment as a fixed effect and baseline value of the parameter under

evaluation as a covariate. Although randomization was stratified by baseline eGFR (≥ 60 vs < 60 mL/min/1.73 m²), this stratification factor was omitted from the final efficacy models due to the small number of participants in the < 60 mL/min/1.73 m² subgroup. Only 31 participants met this criterion (11 in the enlicotide arm, 5 in the bempedoic acid arm, 5 in the ezetimibe arm, and 10 in the bempedoic acid plus ezetimibe arm).

Missing baseline values were imputed with the median of the non-missing baseline values. Missing Day 56 data was handled using a prespecified reference-based imputation strategy, applying a retrieved-dropout approach²⁴ with non-parametric bootstrapping when feasibility criteria were imputed with the washout imputation approach, which assumed participants who prematurely discontinued from the study or who have an intermittent missing Day 56 value would have a ‘washout’ of any potential effect of the assigned study medication. For the washout imputation approach, missing Day 56 values were imputed using data from participants in the active control group who have Day 56 data. Washout imputation was used as the primary analysis for the study as a retrieved dropout approach was not feasible due to the small number of participants with missing data at Day 57 (n=7) (**Supplement Methods**). ApoB and non-HDL-C were analyzed using the same methods applied to LDL-C. All results are reported with two-sided P values and two-sided 95% confidence intervals (CIs). For end points that were not eligible for formal statistical testing, unadjusted 95% CIs are shown. The widths of these 95% confidence intervals have not been adjusted for multiplicity and should not be used to evaluate treatment effects or in place of hypothesis testing.

The percent change from baseline in Lp(a) was analyzed using the aligned rank Wilcoxon test. In this test, the endpoint values are first aligned using the Hodges-Lehmann location shift estimates, and the aligned values are then analyzed using a Wilcoxon rank sum test. The output from this

analysis was used to provide a Hodges Lehmann location-shift estimate of the overall treatment difference with 95% CI. Missing values were multiply imputed using fully conditional specification.

Goal-attainment endpoints were compared using the Miettinen-Nurminen method; participants with missing data were classified as not having achieved the goal.²⁵ All analyses were performed using SAS software, version 9.4 (SAS Institute, Inc.).

Results

Trial Population

Of 435 adults screened, 134 did not meet eligibility criteria (**Figure 1**). The remaining 301 participants (mean age, 64.4 years [SD, 10.1 years]; 37.2% were female) were randomized to enlicitide 20 mg (n = 101), bempedoic acid 180 mg (n = 50), ezetimibe 10 mg (n = 50), or bempedoic acid 180 mg plus ezetimibe 10 mg (n = 100) (**Table 1**). Overall, 298 participants (99.0%) completed the trial (**Figure 1**). The last follow-up visit occurred on March 28, 2025.

Baseline demographics and clinical characteristics were generally balanced across groups (**Table 1**). The mean LDL-C level was 91.6 mg/dL (SD, 30.2 mg/dL). All participants were on statin therapy at baseline (2.3% low-intensity, 50.8% moderate-intensity, and 46.8% high-intensity statin). Overall, 44% of participants had a history of a major ASCVD event, and 56% were at intermediate to high risk for their first major ASCVD event.

The proportion of participants who completed the study was high across the treatment groups: 99% for enlicitide, 96% for bempedoic acid, and 100% for both the ezetimibe and the bempedoic acid plus ezetimibe groups. Adherence to study intervention was also high across groups (mean,

98.2%; range, 97.3%-99.6%). At each visit >95.9% (range across treatment groups: 95.9%-100.0%) of participants reported following administration instructions all or most of the time.

Efficacy

For the primary efficacy endpoint, the mean percentage change from baseline in LDL-C to Day 56 was -64.6% (95% CI, -68.3 to -60.9) in the enlicotide group, -6.3% (95% CI, -13.5 to 0.8) in the bempedoic acid group, -27.8% (95% CI, -32.3 to -23.4) in the ezetimibe group, and -36.5% (-40.8 to -32.2) in the bempedoic acid plus ezetimibe group (**Table 2, Figure 2; Supplementary Table 3**). The corresponding differences in percentage change from baseline between enlicotide versus each comparator were -56.7% (95% CI, -64.3 to -49.0) for bempedoic acid, -36.0% (95% CI, -41.8 to -30.2) for ezetimibe, and -28.1% (95% CI, -33.6 to -22.6) for bempedoic acid plus ezetimibe ($P < 0.001$ for all pairwise comparisons). Maximum LDL-C lowering in the enlicotide group occurred by day 21 and was maintained through day 56 (**Figure 2**). Subgroup analyses suggested that the magnitude of LDL-C-lowering efficacy was consistent across all prespecified demographic and baseline characteristic subgroups. Note that the widths of these 95% confidence intervals have not been adjusted for multiplicity and should not be used to evaluate treatment effects or in place of hypothesis testing (**Supplementary Figure 2**). The individual participant-level distribution of LDL-C changes at day 56 are shown in the waterfall plot (**Figure 2**).

Treatment with enlicotide also resulted in significantly greater reductions in the percentage change from baseline in the key multiplicity-controlled secondary endpoints ApoB and non-HDL-C ($P < 0.001$ for all pairwise comparisons; **Table 2; Supplementary Table 3**). At day 56, the differences between enlicotide versus each comparator for apolipoprotein B were -47.5% (95% CI, -52.9 to -42.1) for bempedoic acid, -33.6% (95% CI, -38.0 to -29.2) for ezetimibe, and -26.8% (95% CI, -31.2 to -22.5) for bempedoic acid plus ezetimibe ($P < 0.001$ for all comparisons). For

non-HDL-C, the differences between enlicitide versus each comparator were -51.2% (95% CI, -57.9 to -44.4) for bempedoic acid, -32.2% (95% CI, -37.1 to -27.2) for ezetimibe, and -25.8% (95% CI, -30.7 to -20.8) for bempedoic acid plus ezetimibe ($P < 0.001$ for all comparisons).

The results for the secondary endpoints that were not multiplicity controlled were as follows. The difference in median percentage change from baseline in Lp(a) was -40.3% (95% CI, -50.5 to -30.0) for enlicitide versus bempedoic acid, -26.2% (95% CI, -33.9 to -18.5) versus ezetimibe, and -41.4% (95% CI, -49.0 to -33.7) versus bempedoic acid plus ezetimibe. At least a 50% LDL-C reduction and an LDL-C level of less than 70 mg/dL was achieved in 81.2% of participants receiving enlicitide versus 2.0% with bempedoic acid (between-group difference: 79.2% [95% CI, 68.4 to 86.3]), 8.0% with ezetimibe (difference: 73.2% [95% CI, 60.0 to 82.0]), and 22.0% for bempedoic acid plus ezetimibe (difference: 59.2% [95% CI, 47.0 to 69.2]). At least a 50% reduction in LDL-C and an LDL-C level of less than 55 mg/dL was achieved in 78.2% of participants receiving enlicitide versus 2.0% with bempedoic acid (difference: 76.2% [95% CI, 65.2 to 83.8]), 8.0% with ezetimibe (difference: 70.2% (95% CI, 56.9 to 79.4), and 20.0% with bempedoic acid plus ezetimibe (difference: 58.2% [95% CI, 45.9 to 68.3]).

Safety

The proportion of participants with at least one AE was 40% with enlicitide, 38% with bempedoic acid, 36% with ezetimibe, and 45% with bempedoic acid plus ezetimibe (**Table 3**). The only AE reported in more than 5% of participants in any group was accidental overdose (defined as taking >1 dose of study medication in a day), all were asymptomatic; none occurred in the enlicitide group. Most common AEs are reported in **Supplementary Table 4**.

Discontinuations due to AEs were infrequent (2% with enlicitide, 4% with bempedoic acid, none with ezetimibe, and 4% with bempedoic acid plus ezetimibe). Serious AEs occurred in four participants in the bempedoic acid group and one participant in the ezetimibe group, with none reported in the enlicitide or bempedoic acid plus ezetimibe groups (**Supplementary Table 5**); one serious AE in the bempedoic acid group was considered drug-related by the investigator. No deaths occurred during the trial.

No participants met protocol-defined criteria for potential drug-induced liver injury or experienced new-onset or worsening diabetes mellitus.

Discussion

In this phase 3, randomized, double-blind, active-comparator trial, oral enlicitide decanoate produced significantly greater reductions in LDL-C than bempedoic acid, ezetimibe, or bempedoic acid plus ezetimibe over 8 weeks of treatment when added to statin therapy. The mean LDL-C change with enlicitide was -64.6%, yielding between-group differences of -56.7%, -36.0%, and -28.1% versus bempedoic acid, ezetimibe, and bempedoic acid plus ezetimibe, respectively. Enlicitide also resulted in statistically significant greater reductions in ApoB and non-HDL-C, as well as greater reductions in Lp(a) and a greater proportion of participants achieving LDL-C goals.

Lipid management guidelines have historically emphasized a stepwise approach to therapy intensification, beginning with maximally tolerated statin therapy followed by the addition of ezetimibe before consideration of other therapies such as PCSK9is.^{5,6} However, this paradigm may contribute to therapeutic inertia, with many patients not receiving timely escalation of lipid-lowering therapy and remaining at risk of preventable CV events.^{26,27} The recent update of the

European Society of Cardiology-European Atherosclerosis Society lipid guidelines recognizes that atherosclerotic CV disease risk is related to cumulative LDL-C exposure and also emphasizes the importance of achieving maximal LDL-C reductions earlier during treatment.^{28,29}

In the present trial, ezetimibe and bempedoic acid, even when used in combination, enabled LDL-C goal attainment in fewer than one-quarter of study participants, whereas enlicitide enabled more than three-quarters of participants to achieve both a reduction of at least 50% in LDL-C and LDL-C levels below guideline-recommended goals, highlighting the potential for enlicitide to address a persistent gap in lipid management that contributes to the ongoing CV disease epidemic.^{2,30}

Beyond LDL-C lowering, enlicitide produced greater reductions in ApoB and non-HDL-C than the oral non-statin comparators, reflecting broader lowering of atherogenic ApoB-containing lipoproteins and total atherogenic particle burden,^{20,21} which are closely linked to residual CV risk.^{29,31} Enlicitide was also associated with modest reductions in Lp(a), a guideline-recognized risk factor^{5,6} that was not reduced by bempedoic acid, ezetimibe, or their combination.³¹

Together, these effects on plasma LDL may translate into greater CV risk reduction compared with the oral comparators evaluated in this trial, whose effects on these parameters were also more modest for ApoB or absent for Lp(a).³¹

This trial has several limitations. The modest sample size (N = 301), and the predominantly older, white study population may limit generalizability of the results. LDL-C reductions with bempedoic acid were lower than expected. No explanation for this attenuated response was identified. Trial conduct, including site footprint and clinical supplies used, baseline characteristics, and discontinuation rates were generally similar between the bempedoic acid alone and the bempedoic acid + ezetimibe arms. Despite this, LDL-C reduction with bempedoic acid alone was lower than expected. One potential cause for this is the known attenuation of

bempedoic acid efficacy when used in conjunction with moderate or high intensity background statin therapy, especially without concomitant use of ezetimibe. Furthermore, the relatively small sample size (50 participants) in this arm may increase the impact of individual participant results on the overall group mean. Notably, LDL-C reductions in the bempedoic acid plus ezetimibe group were consistent with prior studies and enlicitide demonstrated significantly greater efficacy than this combination. Finally, the small sample size and short study duration limit the comparative assessment of safety across treatment arms. However, the safety profiles for ezetimibe and bempedoic acid have been well established in prior studies.^{35,36} Two placebo-controlled, 52-week phase 3 studies, including CORALreef Lipids and CORALreef HeFH, have previously demonstrated the longer-term safety profile of enlicitide.^{20,21} Furthermore, the ongoing CORALreef Outcomes trial (NCT06008756) is evaluating the effect of enlicitide on major adverse CV events in adults at high risk.

In conclusion, over 56 days of treatment, enlicitide demonstrated superior efficacy in lowering LDL-C as well as ApoB, non-HDL-C, and Lp(a) compared with approved oral add-on agents for patients requiring lipid lowering beyond first-line statin therapy. These data support the potential of enlicitide as a new therapeutic option for patients who need lipid lowering beyond what can be provided with statins alone.

Data Sharing: The data sharing policy, including procedures and restrictions, of Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA (MSD), is available at <https://externaldatasharing-msd.com/>.

Figure Legends:

Figure 1: Flow diagram of participants throughout the study. Flow diagram showing the number screened, excluded, randomized, treated, prematurely discontinued, and completed.

Figure 2. Change in Low-Density Lipoprotein Cholesterol (LDL-C) From Baseline to Day 56. A) Waterfall plots; B) Percentage change in LDL-C over time.

Central Illustration. Enlicitide versus Oral Non-Statin Therapies: Day 56 Lipid-Lowering Efficacy

Enlicitide, an oral PCSK9 inhibitor, produced greater reductions from baseline in LDL-C, ApoB, and non-HDL-C to day 56 than bempedoic acid, ezetimibe, or bempedoic acid plus ezetimibe (all $P < 0.001$) in statin-treated adults with a history of a major ASCVD event or at increased risk for a first event.

Abbreviations: ApoB = apolipoprotein B; ASCVD = atherosclerotic cardiovascular disease; LDL-C = low-density lipoprotein cholesterol; non-HDL-C = non-high-density lipoprotein cholesterol; PCSK9 = proprotein convertase subtilisin/kexin type 9.

Trial Registration: ClinicalTrials.gov: NCT06450366

Table 1. Participant Characteristics at Baseline

	Enlicitide 20 mg (n = 101)	Bempedoic acid 180 mg (n = 50)	Ezetimibe 10 mg (n = 50)	Bempedoic acid 180 mg + Ezetimibe 10 mg (n = 100)	Total (N = 301)
Demographics					
Sex, No. (%)					
Male	62 (61.4)	32 (64.0)	32 (64.0)	63 (63.0)	189 (62.8)
Female	39 (38.6)	18 (36.0)	18 (36.0)	37 (37.0)	112 (37.2)
Age, y					
Mean (SD)	64.3 (9.7)	65.4 (10.2)	65.4 (9.3)	63.5 (10.9)	64.4 (10.1)
Median (IQR)	66.0 (58.0-70.0)	64.5 (59.0-74.0)	67.0 (58.0-71.0)	64.5 (58.0-69.5)	66.0 (58.0-71.0)
<65 y, No. (%)	44 (43.6)	25 (50.0)	17 (34.0)	50 (50.0)	136 (45.2)
≥65 y, No. (%)	57 (56.4)	25 (50.0)	33 (66.0)	50 (50.0)	165 (54.8)
<75 y, No. (%)	87 (86.1)	39 (78.0)	43 (86.0)	87 (87.0)	256 (85.0)
≥75 y, No. (%)	14 (13.9)	11 (22.0)	7 (14.0)	13 (13.0)	45 (15.0)
Race, No. (%)*					
White	66 (65.3)	29 (58.0)	33 (66.0)	70 (70.0)	198 (65.8)
Asian	26 (25.7)	16 (32.0)	14 (28.0)	24 (24.0)	80 (26.6)
Black or African American	8 (7.9)	5 (10.0)	3 (6.0)	5 (5.0)	21 (7.0)
Multiple	0	0	0	1 (1.0)	1 (0.3)
American Indian or Alaska Native, White	0	0	0	1 (1.0)	1 (0.3)
Not reported	1 (1.0)	0	0	0	1 (0.3)
Ethnicity, No. (%)[†]					
Hispanic or Latino	25 (24.8)	12 (24.0)	18 (36.0)	28 (28.0)	83 (27.6)
Not Hispanic or Latino	74 (73.3)	38 (76.0)	32 (64.0)	72 (72.0)	216 (71.8)
Not reported	2 (2.0)	0	0	0	2 (0.7)
Geographic Region, No. (%)					
EMEA	28 (27.7)	14 (28.0)	13 (26.0)	25 (25.0)	80 (26.6)
North America	30 (29.7)	17 (34.0)	15 (30.0)	35 (35.0)	97 (32.2)
Latin and South America	21 (20.8)	7 (14.0)	13 (26.0)	23 (23.0)	64 (21.3)
Asia Pacific	22 (21.8)	12 (24.0)	9 (18.0)	17 (17.0)	60 (19.9)
Body mass index					
Mean (SD)	30.1 (6.5)	29.7 (5.7)	29.1 (5.1)	30.6 (6.2)	30.0 (6.0)
Median (IQR)	29.1 (25.4-33.8)	29.2 (25.0-32.8)	28.0 (25.7-32.3)	29.0 (26.2-34.2)	29.0 (25.8-33.4)
<30 kg/m ² , No. (%)	58 (57.4)	29 (58.0)	34 (68.0)	55 (55.0)	176 (58.5)

≥30 kg/m ² , No. (%)	43 (42.6)	21 (42.0)	16 (32.0)	44 (44.0)	124 (41.2)
Estimated glomerular filtration rate, mL/min/1.73 m², No. (%)					
≥30 to <45 mL/min/1.73 m ²	2 (2.0)	0	2 (4.0)	1 (1.0)	5 (1.7)
≥45 to <60 mL/min/1.73 m ²	9 (8.9)	4 (8.0)	3 (6.0)	7 (7.0)	23 (7.6)
≥60 mL/min/1.73 m ²	90 (89.1)	46 (92.0)	45 (90.0)	92 (92.0)	273 (90.7)
Disease characteristics, No. (%)					
Statin intensity[†]					
Low	5 (5.0)	0	1 (2.0)	1 (1.0)	7 (2.3)
Moderate	48 (47.5)	29 (58.0)	24 (48.0)	52 (52.0)	153 (50.8)
High	48 (47.5)	21 (42.0)	25 (50.0)	47 (47.0)	141 (46.8)
ASCVD risk category					
History of major ASCVD event [§]	48 (47.5)	17 (34.0)	26 (52.0)	42 (42.0)	133 (44.2)
Intermediate to high risk for a first major ASCVD event	53 (52.5)	33 (66.0)	24 (48.0)	58 (58.0)	168 (55.8)
History of heterozygous familial hypercholesterolemia	5 (5.0)	3 (6.0)	2 (4.0)	5 (5.0)	15 (5.0)
History of diabetes					
Type 1	1 (1.0)	3 (6.0)	0	1 (1.0)	5 (1.7)
Type 2	44 (43.6)	27 (54.0)	27 (54.0)	51 (51.0)	149 (49.5)
Laboratory values					
LDL-C, mean (SD), mg/dL	89.9 (31.4)	91.9 (24.4)	93.2 (27.4)	92.6 (33.4)	91.6 (30.3)
ApoB, mean (SD), mg/dL	85.0 (22.1)	85.3 (18.0)	84.5 (19.3)	86.4 (25.4)	85.5 (22.1)
Non-HDL-C, mean (SD), mg/dL	114.7 (33.1)	114.2 (26.9)	114.4 (31.5)	115.9 (38.2)	114.9 (33.6)
Lp(a), median (IQR), nmol/L	35.8 (10.9-147.6)	31.7 (12.9-154.5)	32.0 (13.9-121.0)	34.8 (13.9-141.8)	33.9 (12.6-142.6)
Triglycerides					
Median (IQR), mg/dL	116.0 (93.0-156.0)	112.5 (79.0-149.0)	111.5 (88.0-144.0)	118.0 (89.5-154.5)	116.0 (89.0-154.0)
<150 mg/dL, No. (%)	72 (71.3)	38 (76.0)	38 (76.0)	72 (72.0)	220 (73.1)
≥150 mg/dL, No. (%)	29 (28.7)	12 (24.0)	12 (24.0)	28 (28.0)	81 (26.9)

ApoB = apolipoprotein B; ASCVD = atherosclerotic cardiovascular disease; EMEA = Europe, the Middle East, and Africa; HbA1c = hemoglobin A1c; IQR = interquartile range; LDL-C = low-density lipoprotein cholesterol; Lp(a) = lipoprotein(a); n = number of participants randomized to each treatment group; N = total number of participants randomized; non-HDL-C = non-high-density lipoprotein cholesterol; SD = standard deviation.

Conversion factors: To convert LDL-C and non-HDL-C from mg/dL to mmol/L, divide by 38.67; to convert ApoB from mg/dL to g/L, divide by 100; to convert triglycerides from mg/dL to mmol/L, divide by 88.6.

* Self-reported by participants using predefined options; multiple selections were allowed.

[†] Self-reported by participants using predefined options; only one selection was allowed.

[‡] Defined in **Supplement 1**.

^s Defined as having a history of one or more of the following: acute coronary syndrome, myocardial infarction, coronary revascularization, cerebral arterial revascularization, ischemic stroke, or peripheral arterial disease with a history of acute limb ischemia, peripheral arterial revascularization, or major amputation.

^{||} Participants who did not meet the criteria for history of a major ASCVD event at baseline were categorized as being at intermediate to high risk for a first major ASCVD event. Intermediate to high risk for ASCVD event is defined as history of one or more of the following: Possible or definite diagnosis of HeFH based on a locally accepted diagnostic algorithm (eg, AHA algorithm, US MEDPED, Simon Broome, Dutch Lipid Network, or Japanese Atherosclerosis Society Guidelines). Diabetes mellitus. Stable angina pectoris. Transient ischemic attack. Symptomatic peripheral arterial disease without a history of acute limb ischemia, peripheral arterial revascularization, or major amputation. Age greater than or equal to 40 years with ASCVD risk score greater than or equal to 7.5 percent based on the Pooled Cohort Equation or equivalent. Coronary artery calcium score greater than or equal to 100 Agatston units (performed within 3 years of the study).

Table 2. Primary and Secondary Endpoints

	Enlicitide 20 mg (n = 101)	Bempedoic acid 180 mg (n = 50)	Ezetimibe 10 mg (n = 50)	Bempedoic acid 180 mg + Ezetimibe 10 mg (n = 100)
Primary Endpoint (LDL-C):				
Baseline mean (SD), mg/dL	89.9 (31.4)	91.9 (24.4)	93.2 (27.4)	92.6 (33.4)
Day 56 mean (SD), mg/dL	32.0 (20.6)	83.8 (28.0)	67.2 (25.4)	56.3 (21.5)
Mean percentage change from baseline (95% CI) today 56	-64.6 (-68.3 to -60.9)	-6.3 (-13.5 to 0.8)	-27.8 (-32.3 to -23.4)	-36.5 (-40.8 to -32.2)
Between-group difference (95% CI)	--	-56.7 (-64.3 to -49.0)	-36.0 (-41.8 to -30.2)	-28.1 (-33.6 to -22.6)
P value for between-group comparison	--	< 0.001	< 0.001	< 0.001
Key Secondary Endpoints (multiplicity-controlled):				
ApoB				
Baseline mean (SD), mg/dL	85.0 (22.1)	85.3 (18.0)	84.5 (19.3)	86.4 (25.4)
Day 56 mean (SD), mg/dL	38.5 (16.3)	78.9 (17.9)	67.4 (18.1)	60.4 (16.6)
Mean percentage change from baseline (95% CI) today 56	-54.6 (-57.6 to -51.6)	-5.4 (-10.3 to -0.5)	-20.2 (-23.3 to -17.2)	-27.7 (-31.1 to -24.3)
Between-group difference (95% CI)	--	-47.5 (-52.9 to -42.1)	-33.6 (-38.0 to -29.2)	-26.8 (-31.2 to -22.5)
P value for between-group comparison	--	< 0.001	< 0.001	< 0.001
Non-HDL-C				
Baseline mean (SD), mg/dL	114.7 (33.1)	114.2 (26.9)	114.4 (31.5)	115.9 (38.2)
Day 56 mean (SD), mg/dL	48.0 (23.2)	105.7 (30.6)	85.7 (28.8)	75.6 (24.6)
Mean percentage change from baseline (95% CI) today 56	-58.0 (-61.3 to -54.7)	-5.2 (-11.5 to 1.1)	-25.1 (-28.7 to -21.6)	-31.8 (-35.8 to -27.9)
Between-group difference (95% CI)	--	-51.2 (-57.9 to -44.4)	-32.2 (-37.1 to -27.2)	-25.8 (-30.7 to -20.8)
P value for between-group comparison	--	< 0.001	< 0.001	< 0.001
Other Secondary Endpoints (not multiplicity-controlled):				
Lp(a)				
Baseline median (IQR), nmol/L	35.8 (10.9, 147.6)	31.7 (12.9, 154.5)	32.0 (13.9, 121.0)	34.8 (13.9, 141.8)
Day 56 median (IQR), nmol/L	15.9 (6.6, 108.6)	45.7 (16.7, 156.5)	28.6 (12.7, 111.5)	46.6 (15.2, 164.9)
Median percentage change from baseline (IQR) today 56	-26.2 (-43.8, -7.0)	8.1 (-1.7, 30.8)	0.0 (-15.5, 11.8)	10.4 (-2.3, 31.7)
Between-group difference in medians (95% CI)	--	-40.3 (-50.5 to -30.0)	-26.2 (-33.9 to -18.5)	-41.4 (-49.0 to -33.7)
LDL-C Goal Attainment				

Proportion of participants with LDL-C <70 mg/dL and ≥50% reduction from baseline to day 56	81.2	2.0	8.0	22.0
Between-group difference in proportions (95% CI)	--	79.2 (68.4 to 86.3)	73.2 (60.0 to 82.0)	59.2 (47.0 to 69.2)
Proportion of participants with LDL-C <55 mg/dL and ≥50% reduction from baseline to day 56	78.2	2.0	8.0	20.0
Between-group difference in proportions (95% CI)	--	76.2 (65.2 to 83.8)	70.2 (56.9 to 79.4)	58.2 (45.9 to 68.3)

ApoB = apolipoprotein B; CI = confidence interval; IQR = interquartile range; LDL-C = low-density lipoprotein cholesterol; Lp(a) = lipoprotein(a); n = number of participants randomized to each treatment group; Non-HDL-C = non-high-density lipoprotein cholesterol; SD = standard deviation.

Data are presented as mean (SD), median (IQR), or proportion (%), or between-group difference in means or proportions with associated 95% CI and *P* value (for multiplicity-controlled endpoints), as indicated.

Conversion factors: To convert LDL-C and non-HDL-C from mg/dL to mmol/L, divide by 38.67; to convert ApoB from mg/dL to g/L, divide by 100.

No participants had missing baseline LDL-C or non-HDL-C values. One participant in the bempedoic acid arm had missing baseline ApoB and Lp(a) values.

Table 3. Safety Summary*

	No. (%)			
	Enlцитide 20 mg (n = 101)	Bempedoic acid 180 mg (n = 50)	Ezetimibe 10 mg (n = 50)	Bempedoic acid 180 mg + Ezetimibe 10 mg (n = 100)
Adverse events				
Any adverse event	40 (40)	19 (38)	18 (36)	45 (45)
Any moderate or severe adverse event [†]	10 (10)	7 (14)	3 (6)	13 (13)
Discontinued intervention due to an adverse event	2 (2)	2 (4)	0	4 (4)
Serious adverse event [†]	0	4 (8)	1 (2)	0
Drug-related serious adverse event [†]	0	1 (2)	0	0
Died	0	0	0	0

n = number of participants randomized to each treatment group; No. = number of participants with the specified adverse event.

Data are expressed as No. (%), as indicated.

* Adverse events occurred up to and including day 56 after the last dose of study treatment (day 112). Safety set population includes all randomized patients who received at least one dose of study intervention, analyzed according to the treatment administered.

[†] Determined by investigator judgment.

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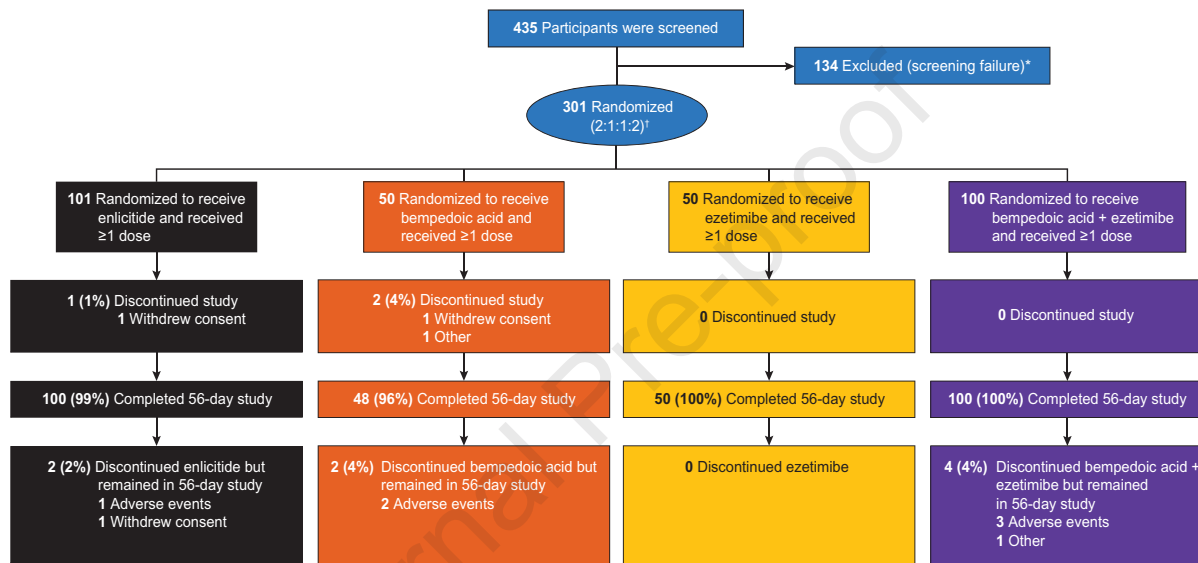
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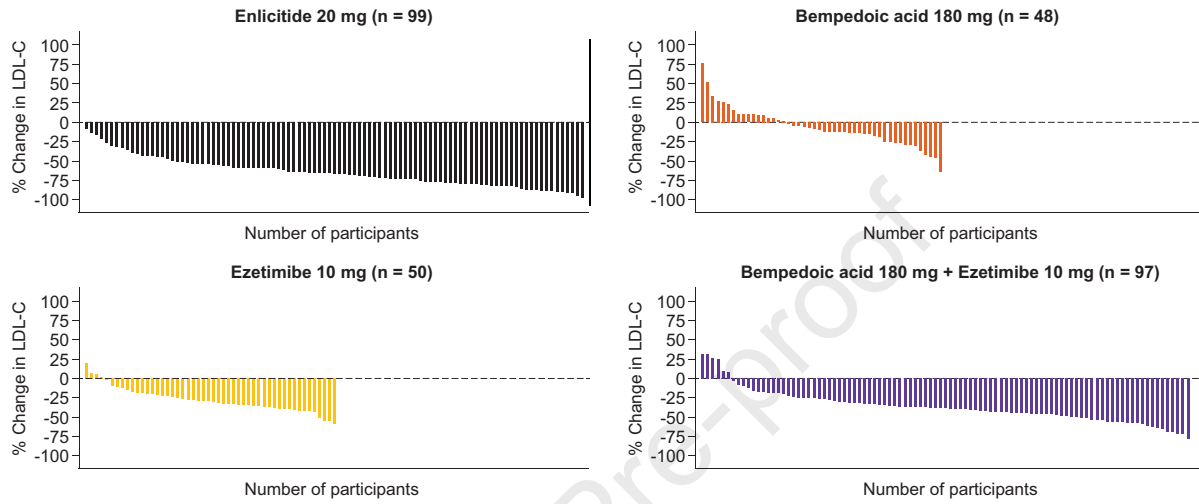
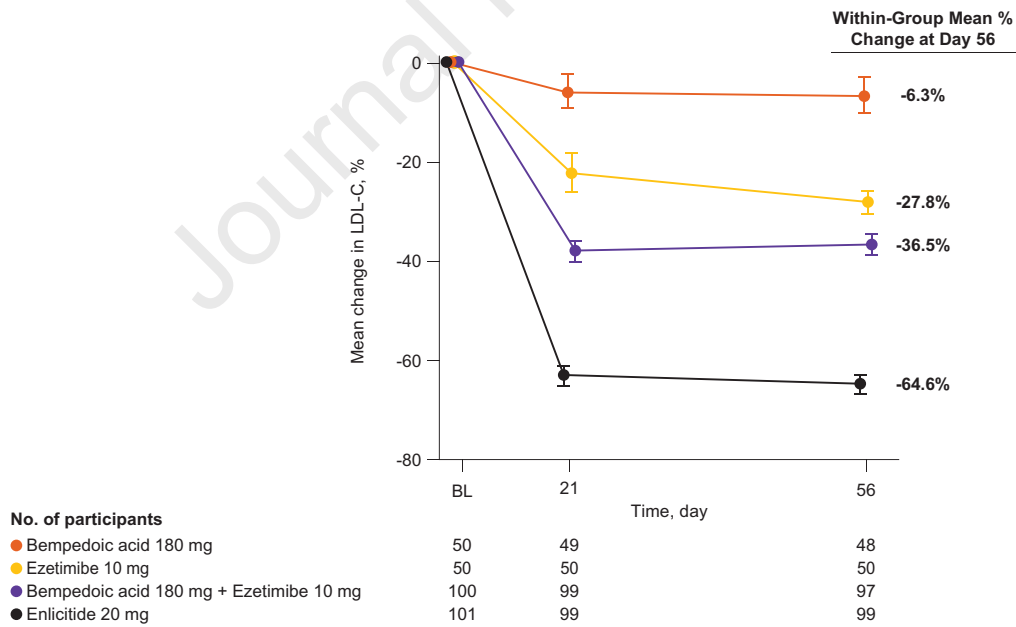
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Figure 1. Flow of Participants in the Trial



* The reasons for screening failure appear in Supplementary Table 2.

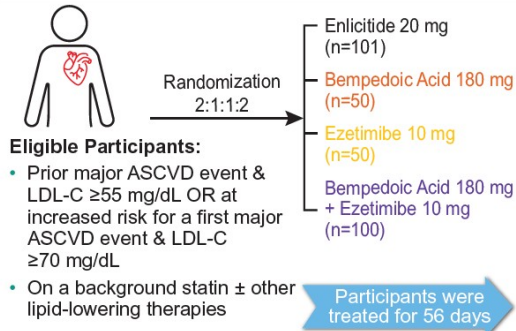
† Randomization was stratified by kidney function (estimated glomerular filtration rate <60 or ≥60 mL/min/1.73 m²) at screening.
ASCVD = atherosclerotic cardiovascular disease.

Figure 2. Change in Low-Density Lipoprotein Cholesterol (LDL-C) From Baseline to Day 56**A. Primary endpoint of percentage change in LDL-C from baseline to 56 days****B. Percentage change in LDL-C from baseline (BL) to 56 days (primary endpoint)**

Clinical Question

Does enlicotide, an oral PCSK9 inhibitor, lead to greater reductions in LDL-C, ApoB, and non-HDL-C than ezetimibe, bempedoic acid, or ezetimibe and bempedoic acid in combination when added to statin therapy?

Study Design



Clinical Implications:

- In statin-treated adults with a history of or at increased risk for a first major ASCVD event, oral enlicotide achieved greater reductions in LDL-C, ApoB, and non-HDL-C at day 56 versus bempedoic acid, ezetimibe, or bempedoic acid + ezetimibe.
- Enlicotide offers an effective treatment option for patients who need additional LDL-C reduction beyond that achieved with statins.

Lipid Lowering Efficacy

